



Location Method of Single-Phase to Ground Fault in Distribution Network Based on Time-Frequency Matrix Analysis of Traveling Wave

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Abstract

This paper proposes a location method based on the analysis of the time-frequency matrix for distribution networks. Firstly, the traveling wave (TW) signal measured by the detection device at the end of each feeder is decoupled to obtain the aerial-mode component of the fault TW. Then, faults are preset at particular locations, such as the main line, primary branch, and secondary branch to construct time-frequency matrices by applying continuous wavelet transform (CWT) to the measured TW signals, and a matrix comparison library is established. The feeder and branch where the fault occurs are determined by comparing the similarity between the matrix extracted from each terminal at the time of fault occurrence and the matrix in the comparison library. Finally, this paper calculates the time for the wave head to reach the measurement point by plotting the energy evolution spectrum of the specific frequency band in the time-frequency matrix, and the precise location of the fault in the distribution network is achieved through the double-ended location method.

Keywords Distribution network · Fault location · Traveling wave · Continuous wavelet transform · Matrix similarity

1 Introduction

The single phase to ground (SPG) fault accounts for more than 70% of all faults in the distribution network and is the most common fault in the distribution network [1, 2]. The fault may lead to voltage surges, causing damage to electrical equipment and, in severe cases, posing a threat to human safety [3]. Therefore, rapid identification and location of faults are crucial [4, 5].

The topology of the distribution network is intricate and variable, and there are many branches on the feeder, which brings great difficulties to the fault location technology. The traditional fault location methods for distribution networks

mainly include impedance methods [6, 7], injection signal methods [8, 9] and TW-based methods [10–12]. The impedance method obtains the line impedance from the fault point voltage and current to determine the fault distance [13]. However, the changes in line parameters, system operating conditions, and transition resistances may affect the accuracy and reliability of the fault location [14–16]. The injection signal method injects characteristic signals into the distribution network and uses signal measurement devices to locate the fault point position [17]. Still, intermittent arcing and transient resistance can affect its reliability [18].

The TW-based method utilizes the propagation and the reflection characteristics of the fault TW signal from the fault point [9]. This method is not affected by transition resistance, voltage/current transformer core saturation, and system oscillation and has been widely applied in the field of fault location in power grids [19].

The TW-based methods can be divided into single-ended methods and double-ended methods according to the principle. The single-ended location method is simple and easy to implement, but the identification of the wave head is relatively difficult. In [20], an improved single-ended location method for identifying the reflected wave head based on the polarity is proposed. However, in the distribution network with complex topology, it is more difficult to distinguish the

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source of wave heads, and the reflected fault TWs are easily overwhelmed by other disturbances or multiple reflected signals [21, 22].

The double-ended method measures the time difference of the initial TWs arrival at two terminals of the line to determine fault distance. It only requires capturing the first arriving wave head, making it easy to identify with high location accuracy [23]. Reference [24] uses the VMD algorithm to decompose the TW signal and calibrate the wave head by using the TEO energy operator. However, the reference needs more research on the relationship between the parameters of TW and frequency. The selection of wave velocity still relies on empirical values, making it challenging to guarantee the precision of location. To address this issue, some scholars choose to conduct time-frequency analysis on TW signals. Reference [25] uses wavelet transform and least squares method to determine the frequency component of the TW that is most suitable for fault location, as well as its corresponding arrival time and propagation velocity, and achieves fault location. Reference [26] uses an improved matrix pencil algorithm to extract the frequency of TW arriving at the bus measuring point, and the corresponding velocity of the incident wave is obtained. However, the above references only extract and analyze a specific characteristic frequency of the TW signal while ignoring the fault information of other frequency bands, which are equally important.

In most TW-based methods, wavelet transform is commonly used to extract and process transient current/voltage signals [27]. However, few scholars have paid attention to the rich fault information contained in the time-frequency matrix obtained after CWT [28]. This paper extensively extracts the fault information by using CWT, presenting multi-dimensional fault data such as time, frequency, amplitude, and polarity in the form of a matrix. Based on the calculation method of vector angles, this paper calculates the similarity between time-frequency matrices to determine the fault branch. An analysis of wave propagation characteristics calculates the wave velocity corresponding to the specific frequency in the time-frequency matrix, thus eliminating the impact of wave velocity uncertainty. The paper uses the double-ended location method to calculate the fault distance; the results demonstrate the accuracy and effectiveness of the proposed approach in fault location under various initial fault conditions, even in some special operating conditions.

Compared with the double-ended location methods discussed before. The main innovation points of this paper are shown below:

(1) By using the generalized Morlet CWT, this paper extensively extracts the rich fault information in the TW signals.

(2) The determination of fault branches in complex structured distribution networks is achieved by fully utilizing the extracted fault information and comparing the matrix similarity.

(3) Calculate the wave velocity corresponding to the selected frequency band, plot the energy spectrum of the specific frequency band in the matrix, and eliminate the influence of wave velocity errors.

The structure of the paper is as follows: [Selection of faulty feeder](#) of the paper presents a comprehensive overview of the construction technique for the time-frequency matrix using CWT. Additionally, it also explains the process of identifying the faulty line based on the constructed matrix. In [Construction of method for fault location](#), we first introduce the basic theory of the double-ended fault location method. Then, the construction of the energy spectrum is presented in detail to extract time information of the wave head. The program of the simulation and discussion is presented in [Experimental analysis](#). The conclusion is presented in [Conclusion](#).

2 Selection of Faulty Feeder

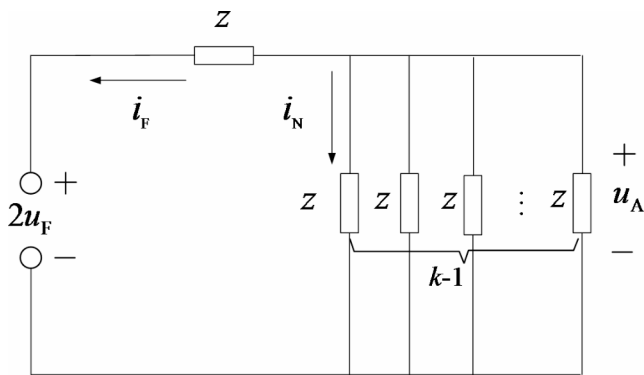
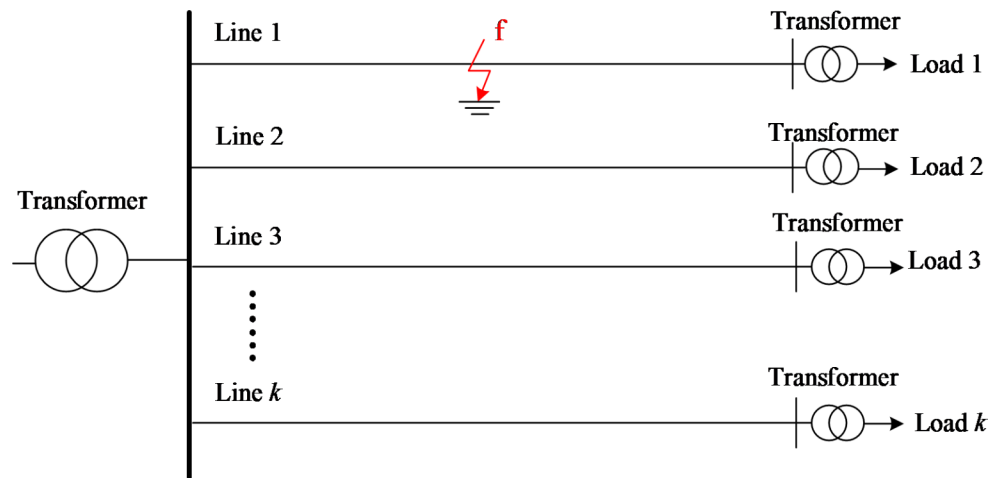
2.1 Characteristics of Fault Traveling Wave

A typical radial distribution network topology model is shown in Fig. 1 The high-voltage bus of the substation connects k distribution network feeders, and each feeder has the same parameters, i.e., the wave impedance is of the same size. Supposing there is a SPG fault at the line 1. According to the wave refraction and reflection description in the Peterson rule [29], we can get an equivalent circuit diagram of the wave propagation process, as shown in Fig. 2.

In Fig. 2, u_F is the voltage TW originating from the fault point, u_A is the voltage TW reaching the bus, i_F is the current TW of the faulty feeder close to the bus, i_N represents the current TW of the feeder other than the faulty feeder close to the bus, and z is the wave impedance of each line. The expression of the amplitude of the TW can be calculated as follows:

$$\begin{cases} u_A = \frac{2}{k} u_F \\ i_F = -\frac{2(k-1)}{k} \frac{u_F}{z} \\ i_N = \frac{2}{k} \frac{u_F}{z} \end{cases} \quad (1)$$

According to the analysis of the formula, the voltage TW measured at the start point of the faulty feeder has the opposite polarity to the current TW. In contrast, the voltage TW measured at other healthy feeders has the same polarity as the current TW. In theory, measuring the polarity of the TW

Fig. 1 Typical radial distribution network model**Fig. 2** Equivalent circuit diagram

enables the identification of the faulty feeder. However, when the value of the grounding resistance is large, or the fault occurs at the moment when the initial phase angle is close to 0° , the amplitude of the voltage TW and the current TW decreases [30], and it is difficult to distinguish the polarity accurately. As a result, it is impossible to determine the faulty feeder accurately.

As there is a one-to-one correspondence between the TW and its fault point [31], the TW contains a large number of characteristic information related to the fault point, such as the time, amplitude, polarity, frequency, etc [32]. Using only one type of specific characteristic information for fault location is incomplete and may result in dead zones. This paper considers utilizing a method to decompose the time-domain TW signals into the time-frequency domain, amplifying rich fault characteristic information.

2.2 Mode Selection of Traveling Wave

When the SPG fault takes place in the distribution network, TWs will be produced from the location of the fault and spread towards every terminal of the feeder, eventually affecting the entire network. Due to the mutual coupling of

electricity and magnetism between the three-phase lines, it is difficult to analyze the propagation characteristic of one particular phase of the TW separately [33]. Therefore, it is necessary to first decouple each phase of the TW before analyzing them. The paper chooses to use the Karen Boolean transform to decouple and merge [34], decomposing the three-phase voltage components u_a , u_b , and u_c into independent aerial-mode (u_α, u_β) and zero-mode components (u_0):

$$\begin{bmatrix} u_0 \\ u_\alpha \\ u_\beta \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} u_a \\ u_b \\ u_c \end{bmatrix} \quad (2)$$

Since the actual line parameters vary with frequency, and there are differences in the propagation paths of each mode component after the Karen Boolean transform, the attenuation and propagation speed of each TW mode component will also be different. Assuming the resistance, inductance, conductance, and capacitance per unit length of the line are denoted as R_0 , L_0 , G_0 , and C_0 , respectively. The parameters related to the TW propagation can be expressed as:

$$\begin{aligned} \gamma(\omega) &= \sqrt{(R_0 + j\omega L_0)(G_0 + j\omega C_0)} \\ &= \alpha(\omega) + j\omega\beta(\omega) \end{aligned} \quad (3)$$

$$A(\omega) = e^{-\gamma x} \quad (4)$$

$$v(\omega) = \frac{\omega}{\beta(\omega)} \quad (5)$$

In the equation above: $A(\omega)$ denotes the transfer function, $\alpha(\omega)$ denotes the attenuation coefficient, $v(\omega)$ denotes the TW propagation velocity, x denotes the propagation distance, and $\beta(\omega)$ denotes the phase coefficient. Based on the line-related model parameters provided in the PSCAD software, the relationship curve between the wave velocity of

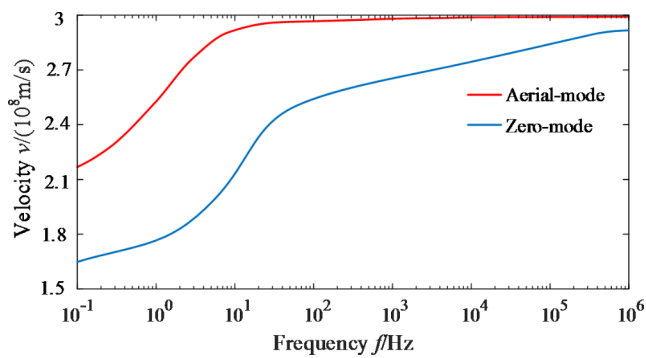


Fig. 3 The velocity of different modes of traveling waves

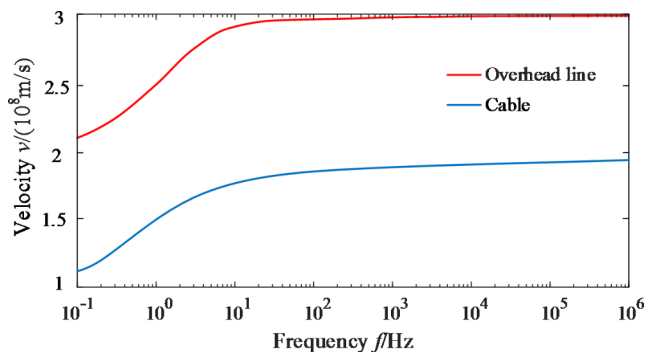


Fig. 4 Traveling wave velocity curves for different types of lines

different modes of TW and frequency can be plotted. Taking the overhead line as an example, the propagation characteristics of different modes of TW in it are illustrated in Fig. 3.

The wave velocity of the TW increases with frequency, and TWs at different frequencies exhibit different velocities. When the frequency increases to a certain value, the wave velocity of the aerial-mode TW tends to stabilize. The ground greatly influences the zero-mode TW, and the relevant parameters of the zero-mode component (such as inductance, wave velocity, resistance, etc.) vary greatly under different frequencies, resulting in significant instability. Compared to the aerial-mode component, the velocity of the zero-mode component shows significant uncertainty within the range of $[0, 10^6]$ Hz. If the zero-mode component is used for location, it will lead to significant errors. Therefore, this paper considers selecting the aerial-mode component of the TW for in-depth analysis.

Due to the generally short feeder lines in the distribution network, the attenuation of the wave velocity v of the aerial-mode TW is weak, so v can be considered as a constant value.

According to the propagation parameters of the TW, Fig. 4, roughly shows the variation curves of the velocity of aerial-mode TW with frequency in different types of lines. The propagation velocity of the aerial-mode component of the TW in the cable line is much lower than that in the overhead line, especially for the high-frequency part.

In addition, whether in the overhead line or cable line, the low-frequency part of the aerial-mode TW undergoes significant changes in wave velocity. However, as the frequency increases, the wave velocity tends to stabilize.

When the frequency reaches the range of $[170, 180]$ kHz, the wave velocities in the cable and overhead line are relatively stable, with a very small range of variation, which can be approximated as constant. According to Eq. (5), the wave velocity of the overhead line in this frequency band can be calculated as 298,000 km/s (v_1), and the wave velocity of the cable line is calculated as 174,500 km/s (v_2). The correspondence curve between frequency and wave velocity in Fig. 4 confirms the accuracy of the wave velocity calculated by this method. Therefore, this paper will analyze the aerial-mode TW signals in the high-frequency range of $[170, 180]$ kHz of the time-frequency matrix in the subsequent fault location study. The velocity (v_1, v_2) of the TW within this frequency band is used as the velocity of the TW propagating throughout the entire network.

2.3 Principle of Continuous Wavelet Transform

As mentioned above, the fault TW includes a series of fault characteristic information, so it is necessary to fuse and thoroughly explore the time-frequency characteristics of the TW. This paper constructs the time-frequency matrix through CWT to achieve the target of extracting the characteristic information contained in the fault TW, achieving fault line selection and location.

The CWT is a versatile time-frequency analysis technique that can dynamically adjust to meet the demands of non-linear signal research through “expansion” and “horizontal shift”. It can focus on any details of the signal [35]. CWT decomposes the aerial-mode component of TW through the wavelet basis function under different scales, generating a series of wavelet coefficients. Assuming that there is a waveform data signal $h(t)$. Applying the CWT to it to obtain the defining equation as follows:

$$C(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} h(t) \psi^*\left(\frac{t-b}{a}\right) dt \quad (6)$$

$$f_{\omega} = \frac{f_c f_s}{a} \quad (7)$$

In Eq. (6), $C(a, b)$ denotes the wavelet coefficient, and $\psi^*(t)$ is the wavelet basis function; a denotes the scale parameter, and b denotes the translation parameter. f_c and f_s are the sampling frequency of the input signal and the center frequency of the wavelet basis function, respectively. The center frequency f_{ω} of different frequency bands can be

obtained by Eq. (7). The result of the CWT is a two-dimensional matrix E :

$$E_{M \times N} = \begin{bmatrix} E_1^1 & E_1^2 & \cdots & E_1^N \\ E_2^1 & E_2^2 & \cdots & E_2^N \\ \vdots & \vdots & \ddots & \vdots \\ E_M^1 & E_M^2 & \cdots & E_M^N \end{bmatrix} \quad (8)$$

M is the number of frequency bands, and N is the number of sampling points under each frequency band divided by the sampling period of the signal. The elements of the matrix are complex numbers. We can obtain the amplitude and phase information of the TW from the elements. Matrix $E_{M \times N}$ represents the magnitude-frequency characteristics of a particular moment through each matrix element in its column vectors while representing the time-domain characteristics of a specific frequency through each matrix element in its row vectors.

After a large number of experiments, the paper found that the Morlet wavelet basis has good symmetry and practicality, so this study uses the Morlet function as the wavelet basis function.

2.4 Establishment of Fault Comparison Library

In order to obtain the sample fault TW signals, faults are preset at specific locations in the distribution network. Construct the time-frequency matrix comparison library by applying CWT to the extracted TW signals. By comparing the similarity of the time-frequency matrices during the occurrence of SPG fault, the branch where the fault occurs can be accurately determined.

Figure 5 is a schematic diagram of a specific 35 kV distribution network. L_1 , L_2 , L_3 , and L_4 are the main lines connecting the 35 kV bus; L_{3-1} , L_{3-2} , and L_{4-1} and L_{4-2} are the primary branches on feeders L_3 and L_4 , respectively. L_{4-1-1} and L_{4-1-2} are the secondary branches on the primary branch L_{4-1} , respectively.

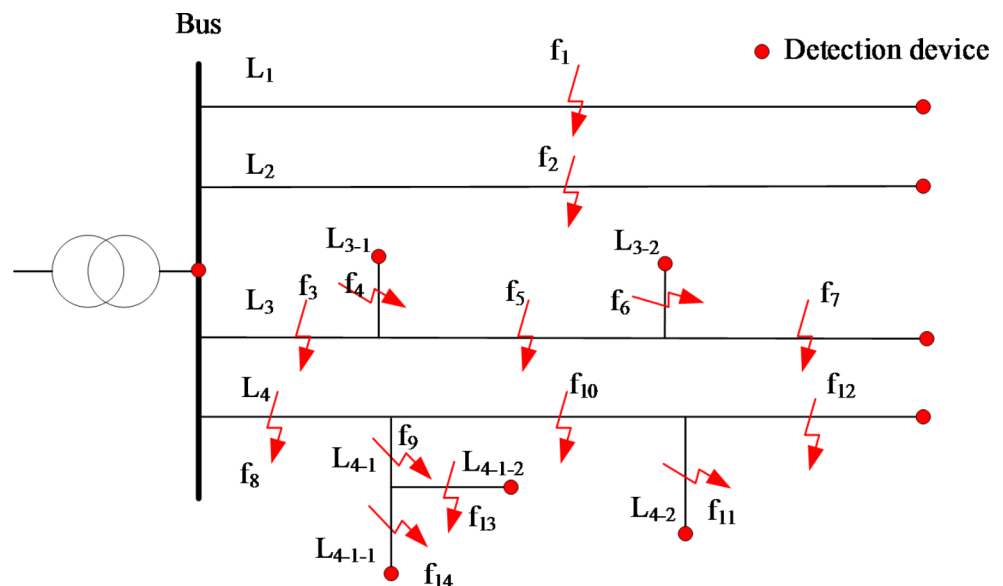
Using the distribution network topology diagram in Fig. 5 as a reference, a simulation model is constructed in PSCAD. TW measurement devices are installed at the terminals of the branches on main lines L_1 , L_2 , L_3 , and L_4 . Then, the preset SPG faults (f_1 to f_{14}) are set at the middle position of each section of the line under three conditions: low grounding resistance, medium grounding resistance, and high grounding resistance, with resistance of 100 Ω , 500 Ω , and 1000 Ω , respectively. Perform CWT on the extracted TW signals to obtain a series of time-frequency matrices and establish a data comparison library for different faulty branches.

2.5 Similarity Calculation of the Matrix

When a SPG fault occurs in the distribution network, the TW is measured by the measurement device installed at the end of each feeder branch. Perform CWT on the measured signal to obtain the time-frequency matrix. Then, the matrix obtained from the measurement points is compared with the matrix in the comparison library to calculate their similarity and determine the branch where the fault is located. This paper utilizes the principle of similarity in image recognition to characterize the similarity between matrices E_1 and E_2 . If matrices E_1 and E_2 have the same number of rows and columns, then we can define the inner product of the matrices as:

$$\langle E_1, E_2 \rangle = \text{tr}(E_2^T, E_1) \quad (9)$$

Fig. 5 Distribution network model



In the Eq. (9), $\text{tr}(\bullet)$ denotes the sum of the main diagonal elements of the matrix. Then, we can calculate the norm of the matrix:

$$\|\mathbf{E}\| = \langle \mathbf{E}, \mathbf{E} \rangle^{\frac{1}{2}} \quad (10)$$

Analogous to the angle between two vectors, the similarity ρ between two matrices can be calculated from Eq. (11):

$$\rho = \frac{\langle \mathbf{E}_1, \mathbf{E}_2 \rangle}{\|\mathbf{E}_1\| \cdot \|\mathbf{E}_2\|} = \frac{\sum_{i=1}^N \sum_{j=1}^M \mathbf{E}(i, j) \mathbf{Q}(i, j)}{\sqrt{\sum_{i=1}^N \sum_{j=1}^M \mathbf{E}^2(i, j) \sum_{i=1}^N \sum_{j=1}^M \mathbf{Q}^2(i, j)}} \quad (11)$$

Among them, $E(i, j)$ is the element of the j th sampling time point at the i th frequency of the time-frequency matrix E in the comparison library. Q is the time-frequency matrix obtained by performing CWT on the measured aerial-mode component of the fault voltage TW. ρ is the similarity coefficient between the two matrices, whose value floats between $[0, 1]$. The two matrices are completely different when $\rho=0$; $\rho=1$ indicates that the similarity between the two matrices is extremely high. The higher the similarity between the matrix measured at the branch and the corresponding time-frequency matrix of the branch in the comparison library, the greater the likelihood of the SPG fault occurring at this branch.

In order to obtain the floating range of the similarity coefficient of the fault time-frequency matrix, the sampling fault points are set at an interval of 25 m at each line in the distribution network model to calculate the similarity coefficient ρ . After conducting extensive simulations and similarity comparisons, we found that when the sampling fault is placed at the branch where the preset fault is located, the similarity between the time-frequency matrices remains above 0.80. Considering special circumstances, such as a fault initial phase angle of 0° or 90° , to retain a certain margin, when the similarity coefficient ρ is greater than 0.85, it is considered that the two matrices are highly similar. We can determine that the fault occurs at the corresponding branch.

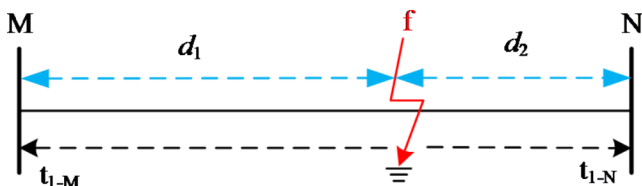


Fig. 6 Schematic diagram of propagation of the TW

3 Construction of Method for Fault Location

3.1 Principle of Double-Ended Location Method

When a SPG fault occurs in the distribution network, fault TW will propagate from the fault location toward both ends of the line. Figure 6 illustrates the relationship between the propagation time of TW and the fault location in the time domain.

In Fig. 6, assuming the SPG fault occurs at position f at the time t_0 , M and N are the TW measurement devices set at both ends of the line, t_{1-M} is the time of the aerial-mode TW reaching device M , and t_{1-N} is the time of the aerial-mode TW reaching device N . The distance from the fault point f to the end M is d_1 , and the distance to the end N is d_2 . The total length of the line is denoted as d_{MN} .

According to the theory of TW propagation, introducing the TW aerial-mode wave velocity v here for distance calculation, the distance from the fault point to both ends can be determined by utilizing the following formula:

$$\begin{cases} d_1 = v(t_{1-M} - t_0) \\ d_2 = v(t_{1-N} - t_0) \end{cases} \quad (12)$$

Combining the condition that $d_{MN} = d_1 + d_2$, eliminate parameter t_0 in Eq. (12):

$$\begin{cases} d_1 = \frac{d_{MN} + v(t_{1-M} - t_{1-N})}{2} \\ d_2 = \frac{d_{MN} + v(t_{1-N} - t_{1-M})}{2} \end{cases} \quad (13)$$

From Eq. (13), we can deduce the conclusion that the principle of the double-ended location method only requires calculating the time of the initial wave head of the TW reaching both ends of the line without the need to calibrate the reflected wave head. This principle is straightforward and very applicable in the real world.

3.2 Fault Location Method Based on Energy Evolution Spectrum

After determining the faulty feeder and branch based on the similarity of the time-frequency matrix, plot the frequency band energy evolution spectrum of $[170 \text{ } 180] \text{ kHz}$ in the time-frequency matrix measured by the device at the terminal of the faulty branch. Then, it can be used to extract the time information of fault, calculate the arrival time of the first wave head.

The initial time for the TW to reach the measurement point is determined by calculating the time corresponding

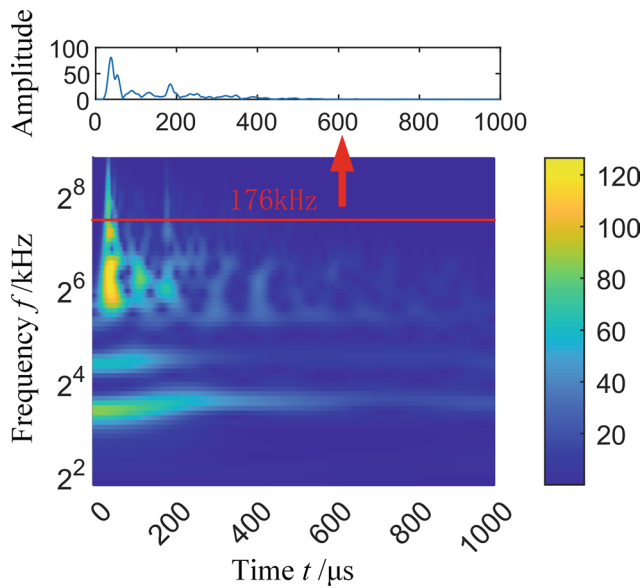


Fig. 7 Matrix measured at the end of line L_{3-1}

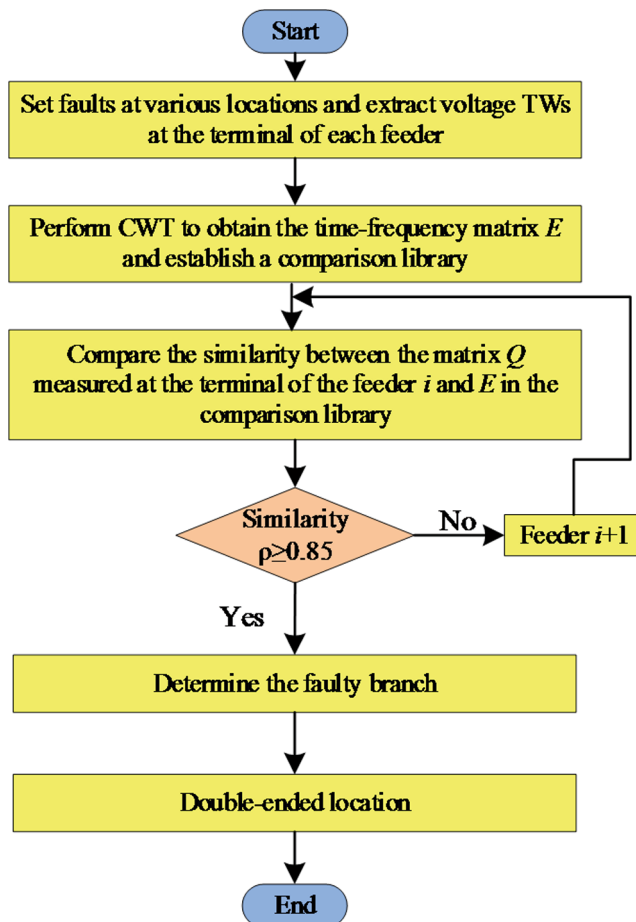


Fig. 8 Flow chart for fault line selection and location

to the first mutation point in the energy evolution spectrum of the signal within this frequency band as shown in Fig. 7.

Based on the calculated wave velocities, the length of the overhead line and cable can be converted, effectively solving the problem of inconsistent wave velocities in mixed lines. The location of the fault can be divided into two situations:

- 1) When the fault occurs at the main line, the time information of the measurement points at both ends of the main line is used for double-ended location.
- 2) When the fault occurs at a particular branch, the time information of the measurement point at the end of the main line and the branch is used for double-ended location.

The flow chart of the line selection and fault location is illustrated in Fig. 8.

4 Experimental Analysis

4.1 Simulation Model Construction

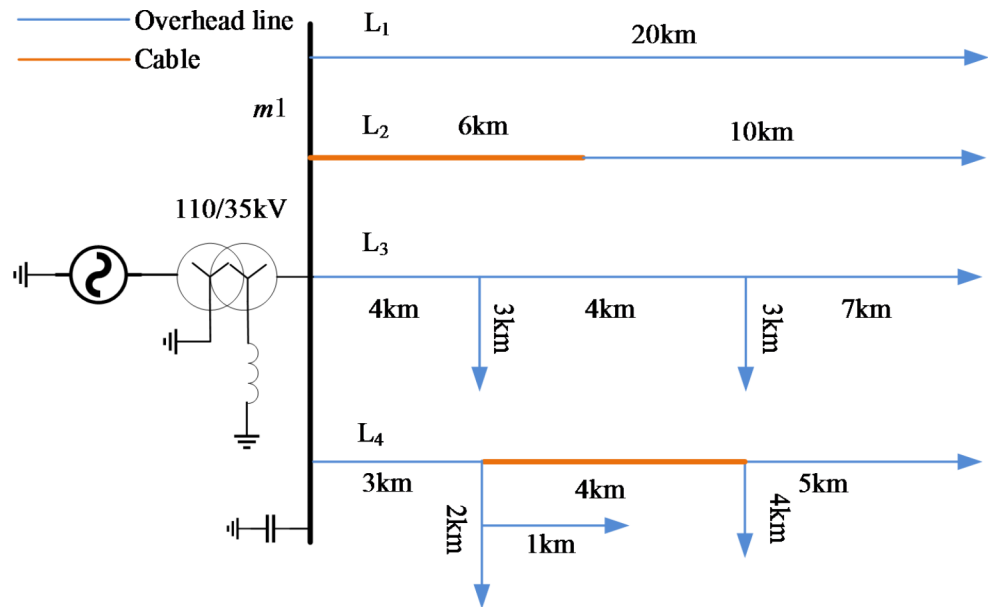
Build a simulation model of the 35 kV distribution network in PSCAD/EMTDC, as shown in Fig. 9. The 35 kV distribution network adopts four different feeders for power supply: single overhead line (L_1), mixed overhead line and cable (L_2), feeder with primary branch (L_3), and feeder with secondary branch (L_4). The transformer ratio is 110/35 kV, and the grounding method of the center point is through an arc suppression coil, with an overcompensation degree of 8%. The load of the line is set as $200 + j12.282\Omega$, and the equivalent capacitance of the bus and circuit breaker is taken as $2\mu\text{F}$. Neglects the mutual effects between the inductance and capacitance; the line parameters of the distribution network are shown in Table 1.

In the simulation, the sampling frequency of the TW measurement device is set as 1 MHz. Considering the calculation speed of CWT and the short length of the distribution network, the refraction and reflection of the TW are complex. Therefore, this paper takes the aerial-mode component of the voltage TW in the 1ms time window for analysis.

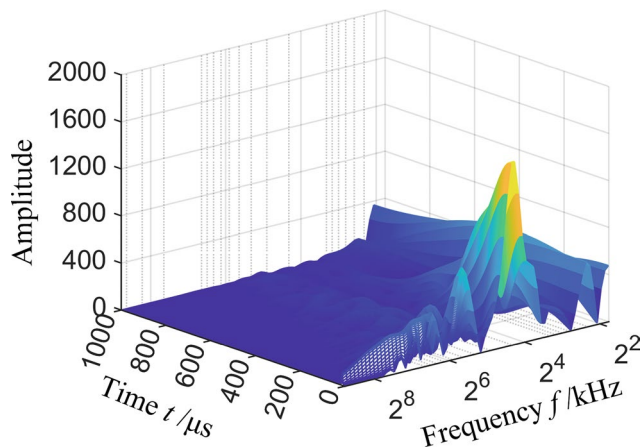
Randomly select the fault point and set the fault initial conditions. Compare the similarity between the matrix of the aerial-mode voltage TW measured at the terminal of each feeder and the data in the comparison library to determine the branch where the fault is located.

4.2 Typical Case Analysis

Assuming that a SPG fault takes place at the main line L_3 , 3.5 km away from the bus. Figures 10 and 11 show the time-frequency matrices of the aerial-mode component of the TW measured at the end of branch L_{3-1} and main line L_3 ,

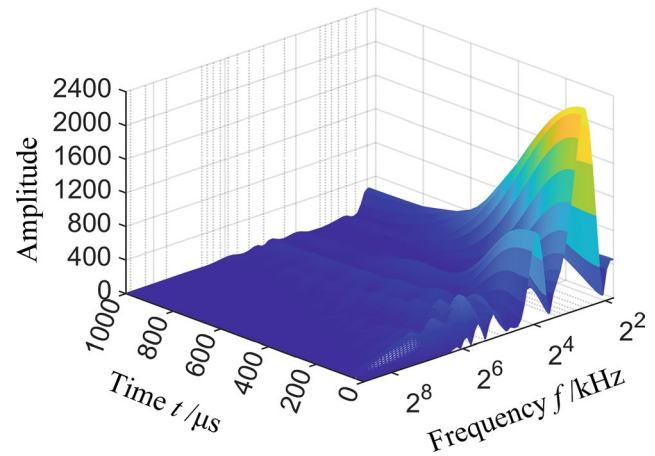
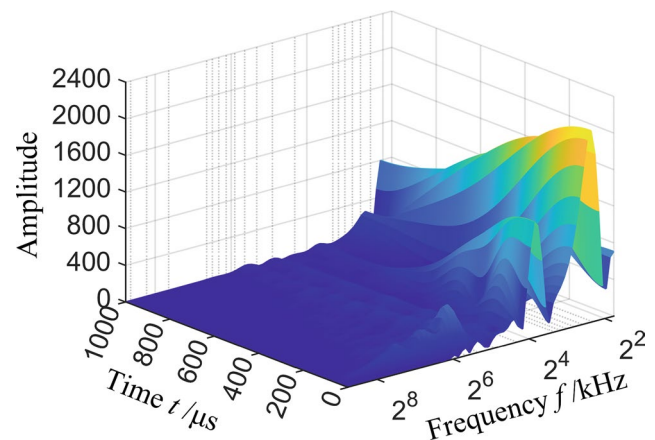
Fig. 9 Simulation model of 35 kV network**Table 1** Parameters of distribution network lines

Parameter		Overhead line	Cable
Resistance (Ω/km)	Positive-sequence	0.172	0.270
	Zero-sequence	0.231	2.721
Inductance (mH/km)	Positive-sequence	1.210	1.019
	Zero-sequence	5.475	0.255
Capacitance ($\mu\text{F}/\text{km}$)	Positive-sequence	0.009	0.339
	Zero-sequence	0.006	0.280

**Fig. 10** Matrix measured at the end of line L_{3-1}

respectively. Figure 12 shows the matrix in the comparison library obtained from the detection point at the end of L_3 when the fault occurs at f_3 .

By comparing Figs. 10 and 11, it is not difficult to find substantial differences in the time-frequency matrices between the healthy line and the faulty line. In the 4 kHz to 10 kHz frequency band, the wavelet coefficient (energy) of the faulty line is significantly greater than that of the healthy line. In the 10–25 kHz range, the energy of the healthy line

**Fig. 11** Matrix measured at the end of line L_3 **Fig. 12** Matrix of fault at f_3 in the library

is greater than that of the faulty line. Meanwhile, Figs. 11 and 12 have relatively similar overall performance. As a result, when calculating the similarity with the matrix in the comparison library, there will be a significant difference in the similarity coefficients between faulty and non-faulty lines, making it easier to distinguish the faulty branch.

Due to the fault occurring at the main line L_3 in this assumption, we should calculate the time for the TW to reread both ends of the main line L_3 using the location method outlined in [Construction of method for fault location](#). Therefore, based on the time-frequency matrices of TW at both ends of the line, the signal energy evolution spectrums within [170,180] kHz are plotted as shown in Fig. 13.

According to the energy evolution spectrum, the time for the fault TW to reach the measurement point is 12 microseconds and 39 microseconds, respectively. The fault point is determined to be located 3.477 km away from the bus

by using the double-ended location formula; the location error is less than 50 m, so the fault location is successfully realized.

4.3 Applicability Analysis

Simulations are conducted in this paper to validate the applicability of the fault location method in this paper. The simulation experiment considers the effects of different initial phase angles of voltage, fault locations, transition resistances, and other special operating conditions. Simulation experiments are carried out by maintaining two parameters and constantly adjusting the remaining single parameter. After extensive simulation analysis, Table 2 presents the line selection and location results of different fault points.

Under the same fault location, different values of transition resistances and initial phase angles are set to analyze

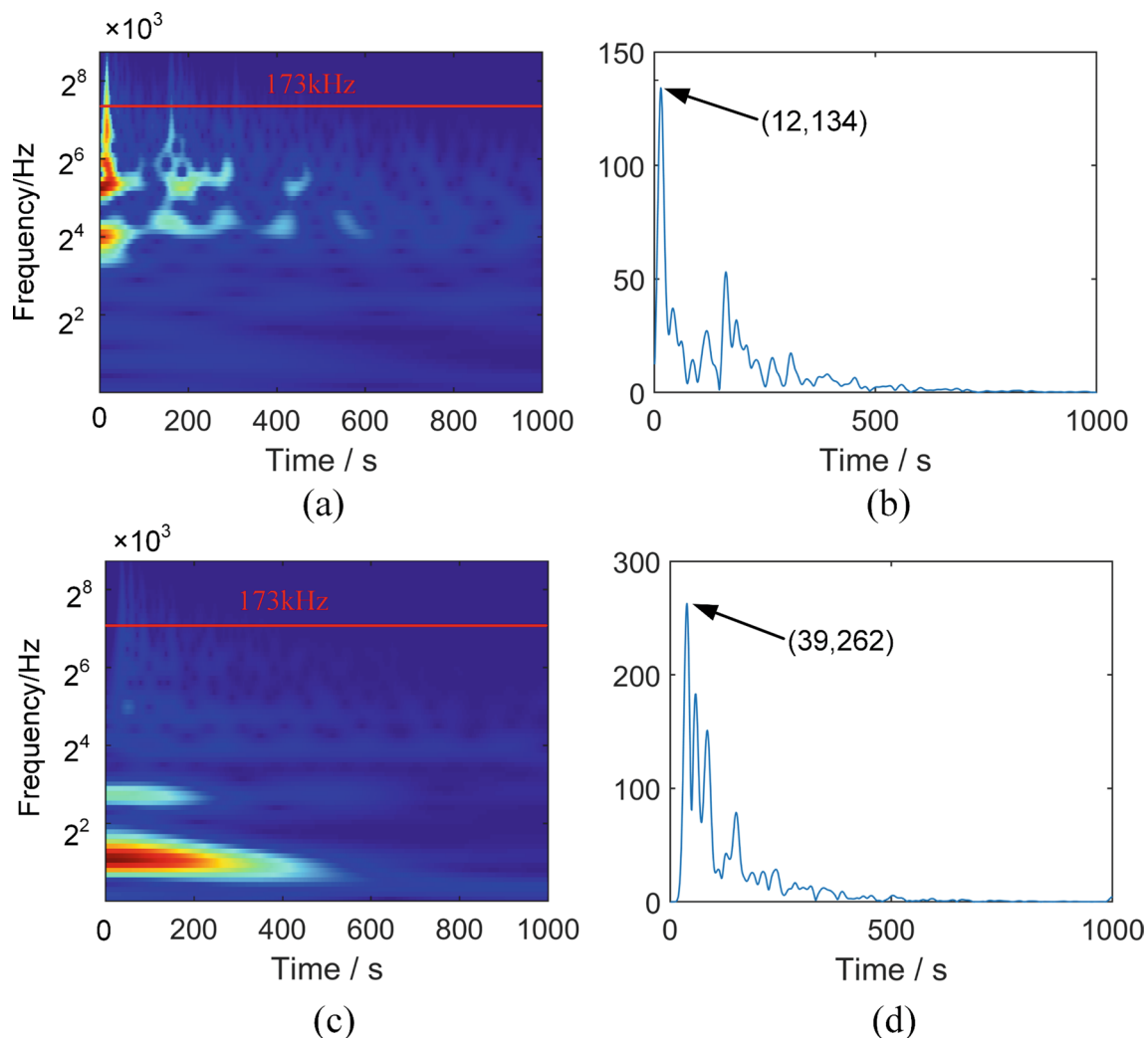


Fig. 13 (a) Time-frequency matrix at the beginning of L_3 (b) Energy evolution spectrum of the matrix in (a). (c) Time-frequency matrix at the end of L_3 . (d) Energy evolution spectrum of the matrix in (c)

Table 2 Results of different fault locations

Fault point	Similarity	Line selection result	Error(m)
L ₁ (5 km)	0.88	L ₁	50.62
L ₃ (10 km)	0.91	L ₃	44.03
L ₃₋₁ (1 km)	0.97	L ₃₋₁	77.20
L ₃₋₂ (2 km)	0.93	L ₃₋₂	38.33
L ₄ (5 km)	0.89	L ₄	59.32
L ₄₋₁ (1 km)	0.92	L ₄₋₁	69.02
L ₄₋₂ (3 km)	0.91	L ₄₋₂	32.86
L ₄₋₁₋₂ (0.6 km)	0.90	L ₄₋₁₋₂	50.55

Table 3 Results of different transition resistances

Fault point	Grounding resistance (Ω)	Similarity	Line selection result	Error(m)
L ₂ (5 km)	200	0.98	L ₂	28.82
	800	0.94	L ₂	48.60
	1000	0.89	L ₂	43.55
L ₃₋₂ (2 km)	200	0.92	L ₃₋₂	36.42
	800	0.89	L ₃₋₂	72.69
	1000	0.92	L ₃₋₂	80.10
L ₄₋₁₋₂ (0.6 km)	200	0.91	L ₄₋₁₋₂	17.43
	800	0.92	L ₄₋₁₋₂	88.38
	1000	0.93	L ₄₋₁₋₂	58.85

Table 4 Results of different initial phase angles

Fault point	Initial phase angle (°)	Similarity	Line selection result	Error(m)
L ₂ (5 km)	0	0.88	L ₂	43.25
	30	0.96	L ₂	52.15
	60	0.91	L ₂	68.50
	90	0.87	L ₂	53.68
L ₃₋₂ (2 km)	0	0.96	L ₃₋₂	62.43
	30	0.93	L ₃₋₂	46.32
	60	0.88	L ₃₋₂	22.42
	90	0.93	L ₃₋₂	28.31
L ₄₋₁₋₂ (0.6 km)	0	0.95	L ₄₋₁₋₂	36.85
	30	0.91	L ₄₋₁₋₂	47.43
	60	0.97	L ₄₋₁₋₂	18.38
	90	0.95	L ₄₋₁₋₂	38.25

their impact on line selection and fault location. Tables 3 and 4 present the obtained results.

When the initial phase angle of the fault voltage is particularly small, the transition resistance is greater than 500 Ω, or the SPG fault occurs at the end of the line, the amplitude of the transient wave mutation is small. After attenuation during the propagation process of the line, the wave head becomes challenging to detect. The simulation results in Tables 2, 3 and 4 indicate that the location method proposed in this paper can accurately locate the SPG fault under different initial conditions. Even if the initial phase angle of the fault is 0° and the transition resistance reaches 1000 Ω, the

Table 5 Results of different grounding methods

Type	Fault point	Similarity	Line selection	Error(m)
High resistance grounding	L ₁ (8 km)	0.96	L ₁	68.82
	L ₃₋₁ (2 km)	0.98	L ₃₋₁	71.60
Arc suppression coil grounding	L ₁ (8 km)	0.91	L ₁	33.55
	L ₃₋₁ (2 km)	0.93	L ₃₋₁	42.12
Ungrounded	L ₁ (8 km)	0.98	L ₁	28.12
	L ₃₋₁ (2 km)	0.99	L ₃₋₁	41.31

maximum location error is only 80 m, indicating high location accuracy. The above experiments demonstrate that this method has good applicability.

Considering the variability of the operating conditions of the distribution network, the following section will examine the adaptability of the proposed method under different network operating conditions.

5 Different Situations of Neutral Grounding Methods

There are three main methods for configuring neutral grounding in a distribution network: high resistance grounding, arc suppression coil grounding, and ungrounded. The different equipment connected to the neutral point will significantly impact the reflection and refraction of the TW reaching the bus, so an in-depth analysis is needed. Set the grounding points at the line where f_1 and f_4 are located, with the grounding resistance of 300 Ω and an initial phase angle of 15°. Table 5 demonstrates the obtained location results.

6 Situations of Different Capacitances in Parallel with the Bus

Generally, reactive power is compensated on-site by connecting capacitors in parallel with the busbar to compensate for the required reactive power and to maintain voltage stability at the busbar. The value of the parallel capacitance will affect the system's resonant frequency, thereby affecting the high-frequency component of the TW. Table 6 shows the line selection and fault location results of connecting different capacitors in parallel with the bus.

The fault location results in Tables 5 and 6 clearly show that the method introduced in this study can still accurately select the faulty branch and locate the SPG fault under some unique operating conditions. The location error is generally within 100 m and has good robustness.

Table 6 Results of connecting different capacitors in parallel with the bus

Parallel capacitance value (μF)	Fault point	Similarity	Line selection	Error(m)
1	L_1 (8 km)	0.92	L_1	54.32
	L_{3-1} (2 km)	0.87	L_{3-1}	28.31
2	L_1 (8 km)	0.93	L_1	78.27
	L_{3-1} (2 km)	0.93	L_{3-1}	26.21
2.5	L_1 (8 km)	0.94	L_1	64.12
	L_{3-1} (2 km)	0.89	L_{3-1}	56.24

Table 7 Comparison of results of different location methods

Fault point	Error of reference [25]/m	Error of reference [27]/m	Error of this paper/m
L_1 (8 km)	104.68	86.24	23.78
L_2 (5 km)	125.32	74.32	36.43
L_{3-1} (2 km)	125.32	86.36	52.06
L_4 (5 km)	96.87	95.63	25.42

6.1 Comparison to Existing Methods and Discussions

To verify the superiority of the location method proposed in this paper, it is compared with the fault location methods proposed in references [24] and [26]. Four SPG faults are simulated and the transition resistance is set as 100Ω . The comparison of location results is shown in Table 7.

Table 7 shows that the network location method in [24] considers the initial wave arrival times from all measurement points, thus avoiding the impact of time errors from some measurement points. However, if the wave velocity value selected by this method differs significantly from the actual wave velocity, the location accuracy is still not high. In contrast [26], considers the relationship between TW velocity and frequency. However, the location results still exhibit certain errors due to the relatively subjective selection of frequency bands. This paper constructs a multidimensional information matrix and plots the energy evolution spectrum of the specific frequency corresponding to the wave velocity, thereby optimizing the calculation of TW velocity and avoiding the influence of TW velocity errors. The maximum absolute location error is 52 m, demonstrating higher location accuracy.

7 Conclusion

On the basis of the principle of the uniqueness of TW time-frequency waveforms, this paper uses the time-frequency matrix of the voltage aerial-mode TW for SPG faults and the corresponding relationship between the fault branches to form a criterion. By building a comparison library, the

fault branch is determined. Then, the arrival time of the wave head will be calibrated based on the information in the time-frequency matrix to achieve the accurate location of SPG faults in the distribution network. After extensive simulation and calculation, the following conclusion has been drawn:

(1) The voltage TW during SPG fault contains rich fault characteristic information. The time-frequency matrix obtained after CWT contains the amplitude, polarity, phase, and frequency characteristics of the TW. By calculating the similarity between time-frequency matrices, the fault characteristics of TW can be utilized to a greater extent, achieving accurate determination of fault branch.

(2) Plotting the energy evolution spectrum in specific frequency band of the time-frequency matrix. From this, it is possible to accurately calibrate the time it takes for the first wave head of the TW to reach the detection point. Combined with the wave velocity determined by specific frequency, precise location of SPG fault can be achieved.

(3) This line selection and location method is not affected by factors such as grounding resistance, initial phase angle, neutral point grounding method, etc., and maximizes the use of fault information of TW with strong adaptability.

The paper will consider reducing the number of measurement devices installed in the network without decreasing the fault location accuracy in the future, making the method more economical.

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Declarations

Competing interests The authors declare that they have no competing interests.

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